M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 10 (10TH MATRIC) - CHEMISTRY

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks (Theory: 60 | Practical: 15) | Time: 2 Hours 45 Minutes

SECTION 1: BISE GUJRANWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- **Section A: Objective (MCQs):**
  12 Compulsory MCQs (12 Marks).
- **Section B: Short Questions:**
  Attempt 15 out of 24 Short Questions (Q2, Q3, Q4: 5/8 each = 30 Marks).
- **Section C: Long Questions:**
  Attempt 2 out of 3 Long Questions (9 Marks each = 18 Marks).

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Chapter 9: Chemical Equilibrium:

Reversible vs Irreversible reactions, Dynamic equilibrium, Law of Mass Action (Kc = [products]/[reactants]), Units of Kc, Equilibrium constant importance (direction and extent of reaction).

Chapter 10: Acids, Bases and Salts:

Concepts of Acids & Bases: Arrhenius, Bronsted-Lowry (Conjugate acid-base pairs), Lewis (Electron pair donor/acceptor), Self-ionization of water (Kw = 1.0 x 10^-14 at 25 deg C), pH and pOH scale, Indicators, Salt preparation methods.

Chapter 11: Organic Chemistry:

Organic compounds definition, Vital Force Theory and Wohler synthesis of urea, Characteristics of organic compounds, Representation formulas (Molecular, Structural, Condensed, Dot & Cross), Functional Groups (Alcohol, Aldehyde, Ketone, Carboxylic acid, Ester, Amine), Isomerism.

Chapter 12: Hydrocarbons:

Alkanes (Saturated, Combustion, Halogenation substitution), Alkenes (Unsaturated, Dehydration of alcohols, Hydrogenation, Bromine water test), Alkynes (Unsaturated, Preparation of ethyne, Baeyer test, Acidity of alkynes).

Chapter 13: Biochemistry:

Carbohydrates (Monosaccharides, Oligosaccharides, Polysaccharides), Proteins (Amino acids, Peptide bond), Lipids (Fatty acids, Triglycerides), Nucleic Acids (DNA double helix model, RNA), Vitamins (Fat-soluble A, D, E, K vs Water-soluble B, C).

Chapter 14: Environmental Chemistry I: Atmosphere:

Layers of atmosphere (Troposphere, Stratosphere, Mesosphere, Thermosphere), Major Air Pollutants (CO, CO2, SO2, NOx), Greenhouse effect and Global Warming, Acid Rain formation and effects, Ozone depletion in Stratosphere by CFCs.

Chapter 15: Environmental Chemistry II: Water:

Properties of water, Soft vs Hard water, Causes of hardness (Calcium and Magnesium salts), Removing temporary (boiling, Clark method) and permanent hardness (washing soda, ion-exchange resin), Water pollution and waterborne diseases (Cholera, Typhoid, Hepatitis).

Chapter 16: Chemical Industries:

Basic metallurgical operations (Crushing, Concentration, Froth flotation, Roasting, Smelting, Bessemerization), Solvay Process for sodium carbonate (Flowsheet & reactions), Urea manufacture (Ammonia synthesis, Carbon dioxide reaction, Granulation), Petroleum refining fractions.

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

*** Write balanced chemical equations with state symbols for all Solvay process reactions and organic halogenations.**

*** Clearly define Bronsted-Lowry conjugate acid-base pairs with direct reaction examples.**